

# The Effects of Expectation: Evidence from Text-Based Infectious Disease Data

Ram Joshi      Xiaowen Lei      Julian F. Ludwig      Xiaohan Ma \*

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## Abstract

This paper investigates the causal impact of expectations on real economic outcomes by developing a novel identification strategy that isolates the effect of expectations on the macroeconomy. Our approach avoids the need to make specific assumptions about how expectations interact with business cycles or how these expectations are generated. Using newspaper coverage of infectious disease outbreaks in the United States as an instrumental variable, we capture shifts in economic agents' expectations that are unrelated to actual economic fundamentals. Applying this method within a Structural VAR framework, we find that a 1% decline in GDP forecasts can lead to a reduction in real GDP by up to 0.49%. These results provide clear evidence that expectations do influence macroeconomic outcomes, supporting a wide body of research that highlights the importance of expectations in shaping economic dynamics.

*Keywords:* Expectation, Information Frictions, Newspaper Data, Instrumental Variable, SVAR, Business Cycle

*JEL Classification Codes:* C36, D84, E32.

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\*Ram Joshi is from the Division of Business and Mathematics at Blinn College, Bryan Campus. Xiaowen Lei is from the Department of Economics and Finance, University of Guelph. Julian F. Ludwig and Xiaohan Ma are from the Department of Economics at Texas Tech University.

# 1 Introduction

Expectations are a cornerstone of macroeconomic dynamics, shaping how individuals and firms make decisions about consumption, investment, and policy. They serve as the crucial link between current economic conditions and future outcomes, influencing fluctuations across the broader economy. Yet, identifying the causal impact of expectations on macroeconomic variables remains one of the most challenging tasks in empirical economics. As highlighted by [Lucas Jr \(1976\)](#), this difficulty arises from the complex interplay between policy, expectations, and economic outcomes. When economic agents form expectations based on anticipated policy changes, it becomes nearly impossible to disentangle cause and effect using observational data alone. Are observed economic fluctuations the result of policy shifts, or are they driven by agents' expectations about future changes? This identification problem, deeply rooted in the empirical macroeconomic literature, makes it particularly challenging to quantify how much of observed behavior is forward-looking versus backward-looking.

Traditional full-information rational expectations (FIRE) models<sup>1</sup> incorporate expectations as a key determinant of economic outcomes, with their empirical performance assessed *indirectly* through model-implied equilibrium relationships under maintained assumptions. Because expectations are endogenously linked to observed fundamentals in these frameworks, they are not separately identified in the data. For example, in estimating consumption habit models, both structural habit-formation models and reduced-form Euler-equation estimates attribute persistence to habits under maintained assumptions about expectations, typically ruling out independent belief shocks. In contrast, our approach exploits external instruments to generate exogenous variation in observed expectations, allowing us to *directly* estimate their causal effects without imposing a particular structural mechanism a

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<sup>1</sup>We follow the definition of full-information rational expectations (FIRE) where agents observe all payoff-relevant fundamentals and form rational expectations conditional on that information set.

priori.

In this paper, we ask a simple yet profound question: to what extent do shifts in economic agents' expectations, independent of realized economic fundamentals, influence macroeconomic outcomes? We address this challenge by explicitly demonstrating the identification problem and arguing that an instrumental variable structural vector autoregressive (IV-SVAR) approach is needed to overcome it. Conceptually, an ideal experiment would involve inducing changes in beliefs without altering fundamentals: if expectations play a key role, such belief shifts should temporarily affect economic decisions and aggregate outcomes until the underlying information is revealed. News about outbreaks of infectious diseases, such as MERS, SARS, Ebola, H5N1, or H1N1, closely mirrors this thought experiment. When these diseases do not materialize into full-scale health crises, they do not directly disrupt production or labor supply, implying that any observed economic effects operate primarily through changes in expectations. At the same time, infectious-disease news is inherently salient because, if such risks were to materialize, they could trigger severe economic consequences, including production disruptions, labor supply constraints, regulatory shutdowns, and demand shortfalls. This combination of limited immediate realization and potentially catastrophic downside risk makes infectious-disease news a powerful driver of uncertainty and pessimism about future economic activity, and therefore a natural source of exogenous variation in output expectations.

We use infectious disease events as instruments to influence economic agents' expectations about GDP, measured through survey-based GDP forecasts, and examine how these shifts in expectations impact the broader economy. Our empirical analysis focuses on the United States, combining U.S. newspaper-based measures of infectious disease coverage with survey-based GDP forecasts and aggregate macroeconomic data. Our findings provide robust evidence that expectations have significant and measurable effects on macroeconomic outcomes. We show that a 1% decline in GDP forecasts due to infectious disease scares leads to a reduction in real GDP by up to 0.49%, indicating a 50% pass-through from expectations to actual output. This substantial effect highlights the powerful role of expectations

in driving economic fluctuations. Additionally, we observe that these expectation-driven impacts extend beyond GDP, with significant declines in stock prices, interest rates, and the price level following negative shifts in expectations.

The effects we document are transitory, with economic variables gradually returning to their baseline level as expectations adjust back to align with actual fundamentals. Our results suggest that the persistence of these effects depends on the speed at which agents update their expectations. We further validate our findings through a series of robustness checks. Using alternative instruments, such as the Infectious Disease Frequency (IDF) index, and incorporating different measures of economic uncertainty, including the VIX and macro uncertainty indices, we demonstrate that our core results hold across a range of model specifications. Moreover, we compare our IV-SVAR approach to alternative estimation methods, such as instrumental-variable local projections, and find consistent results, underscoring the reliability of our identification strategy.

Our proposed approach contributes to the literature by providing direct empirical evidence on the role of expectations in driving economic fluctuations. In existing literature, expectations either affect outcomes by construction within a given model, e.g.: full-information rational expectations (FIRE) models, or are identified within VARs using exclusion, zero, or sign restrictions that depend on a tightly specified joint data-generating process. The instrumental variable approach we adopt here does not require strong *a priori* restrictions on the nature of the events that drive or form expectations or the dynamic responses of economic agents to their expectations, both of which could potentially have significant impacts on the ultimate outcomes. Instead, our instrumental variable method can serve as a powerful tool to empirically validate the key assumptions in those models.

The organization of this work is as follows: Section 3 explores the model and outlines the necessity for IV estimation. Section 4 introduces the data and details the model estimation. We present the main findings and robustness checks in Section 5. Section 6 concludes.

## 2 Literature Review

Since [Lucas Jr \(1976\)](#)’ critique emphasized the joint determination of policy, expectations, and outcomes, early macroeconomic theory formalized this interdependence using models with full-information rational expectations (FIRE). In these canonical FIRE models, aggregate fluctuations are driven primarily by fundamental shocks, such as to technology or preferences, while expectations efficiently incorporate these shocks and typically do not constitute an independent source of dynamics. Consequently, subsequent theoretical work has explored expectation-driven fluctuations by relaxing the FIRE assumption, allowing expectations to deviate from fundamentals through learning, information frictions, or behavioral mechanisms. [Evans and Honkapohja \(2001\)](#) summarizes how learning generates persistent belief dynamics and policy non-neutralities even in otherwise standard macro models. Sticky information models [Mankiw and Reis \(2002\)](#) generate inertia through infrequent updating, noisy-information frameworks [Woodford \(2003\)](#) emphasize imperfect signal extraction. Rational inattention [Sims \(2003\)](#); [Maćkowiak and Wiederholt \(2009\)](#) endogenizes limited attention as an optimal choice, while [Hansen and Sargent \(2011\)](#) explores decision rules when agents confront model uncertainty and [Bordalo, Gennaioli and Shleifer \(2018\)](#) examines the effect of diagnostic expectations on business cycles. A complementary line of research studies news and belief shocks, emphasizing how advance information about future fundamentals can affect current economic activity ([Beaudry and Portier, 2004, 2006](#)). Subsequent contributions deepen this perspective by clarifying the role of information quality and structure in shaping the propagation of news and belief-driven fluctuations ([Blanchard, L’Huillier and Lorenzoni, 2013](#); [Chahrour and Jurado, 2018](#); [Beaudry and Portier, 2014](#)). A distinct literature formalizes sentiment shocks as extrinsic disturbances to higher-order beliefs that can drive fluctuations even in the absence of fundamental news (See [Angeletos and La’o \(2013\)](#)). Finally, behavioral macro models allow non-rational belief formation directly, generating self-reinforcing waves of optimism and pessimism (“animal spirits”) through rule-switching (See [De Grauwe \(2012\)](#)) or higher-order belief coordination under

dispersed information [Angeletos and La’o \(2013\)](#). Against this backdrop, our IV approach provides reduced-form causal evidence on the macroeconomic effects of exogenous shifts in observed expectations, without committing to a specific learning, information, or behavioral mechanism.

Motivated by these theoretical developments, a growing empirical literature seeks to quantify the role of expectations in driving business-cycle fluctuations using a range of identification strategies. [Chahrour, Nimark and Pitschner \(2021\)](#) study misinformation in a structural model, modeling media-driven belief distortions as part of agents’ information set. [Assenza et al. \(2024\)](#) use fake technology news as an instrument for macroeconomic uncertainty within a proxy-VAR framework. [Benhima and Poilly \(2021\)](#) estimate a structural model in which demand fluctuations arise from misperceptions about fundamentals via noisy signals. [Enders, Kleemann and Müller \(2021\)](#) identify belief shocks through a statistical decomposition of professional forecast errors. At the micro level, there is evidence that firm expectations ([Zeno Enders, 2022](#)) and household expectations ([Roth and Wohlfart, 2020](#)) significantly impact economic activity. In financial markets, survey evidence confirms that beliefs affect asset prices and portfolios ([Giglio et al., 2021](#)). Finally, [Haaland, Roth and Wohlfart \(2023\)](#) uses experimental methods to study the formation of economic beliefs.

Within this empirical literature, the paper most closely related to ours is [Lagerborg, Pappa and Ravn \(2023\)](#), which uses mass-shooting events as an instrument for economic expectations, interpreting their effects as shifts in pessimistic sentiment that operate without direct aggregate economic impact. A key difference is that mass-shooting events are unlikely to have direct aggregate economic consequences, whereas infectious disease risk would have substantial macroeconomic effects if realized. By focusing on episodes in which disease risk does not materialize, our approach isolates expectation-driven effects while allowing expectation changes to be fully rational responses to genuine economic risk.

### 3 Model

In this section, we first show two identification problems associated with estimating a linear regression model with expectation variables, which explains the necessity of using an instrument to measure the effect of expectations. We then discuss the key assumptions for our approach and present a univariate model and a structural vector autoregression (SVAR) model augmented with an instrumental variable (IV) to estimate the impulse response functions (IRFs) to an exogenous change in expectations.

#### 3.1 Identification Issues

Consider a time series variable  $Y_t$  whose dynamics (data generating process) is given by the following equation:

$$Y_t = \delta + \gamma Y_{t-1} + \phi Y_{t+1}^e + \epsilon_t \quad (1)$$

where  $\epsilon_t \stackrel{i.i.d.}{\sim} N(0, \sigma_\epsilon^2)$  represents an independent shock,  $\gamma$  captures the dependence of  $Y_t$  on its previous value, and  $\phi$  quantifies the dependence of  $Y_t$  on its future expectations  $Y_{t+1}$ .

The objects of interest are  $\phi$ , the impact of expected  $Y_{t+1}$  on current  $Y_t$ , and their subsequent effects on future  $Y_{t+h}$ , influenced by both  $\phi$  and  $\gamma$ . We assume data is available for both  $Y_t$  and  $Y_{t+1}^e$ . However, despite the availability of this data, two identification problems prevent us from estimating (1) with a linear regression:

The first problem arises from the common assumption of full information and rational expectations (FIRE), which implies the following equation for expectations:

$$Y_{t+1}^e = b_0 + b_1 Y_t \quad (2)$$

where  $b_0 = \frac{\delta + \phi b_0}{1 - \phi b_1}$  and  $b_1 = \frac{\gamma}{1 - \phi b_1}$ .<sup>2</sup> Equation (2) reveals that under FIRE,  $Y_{t+1}^e$  is a linear function of  $Y_t$  only, and therefore does not provide the variation needed to identify the coefficients  $\gamma$  and  $\phi$  in (1), summarized as:

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<sup>2</sup>See Appendix A for derivation.

**Issue 1.** *Under rational expectations, i.e.,  $Y_{t+1}^e = E[Y_{t+1}|\mathcal{F}_t]$ , and full information, i.e.,  $E[\epsilon_{t+h}|\mathcal{F}_t] = 0$  for  $h > 0$  and  $E[\epsilon_{t+h}|\mathcal{F}_t] = \epsilon_{t+h}$  for  $h \leq 0$ , equation (1) cannot be identified.*<sup>3</sup>

That is, a central identification challenge is the difficulty of disentangling the causal impact of exogenous changes in expectations on aggregate economic outcomes. This difficulty arises because expectations, realizations, and current decisions are jointly determined. Recent advances, summarized in [Bachmann, Topa and van der Klaauw \(2022\)](#), have substantially improved the measurement of household and firm expectations and integrated these measures into empirical and structural analysis. However, existing work generally do not propose or implement research designs that *exogenously* shift expectations to isolate their causal effect on aggregate outcomes. Our contribution is to address this challenge by adopting an instrumental-variables (IV) approach that exploits exogenous variation in observed expectations, allowing us to estimate their direct causal impact on macroeconomic activity.

Issue 1 can be resolved if  $Y_{t+1}^e$  can occasionally move independently of  $Y_t$ , which necessitates a departure from FIRE. However, the following identification problem remains:

**Issue 2** (Simultaneity). *If  $Y_t$  is perceived persistent, i.e.,  $Y_{t+1}^e$  depends on  $Y_t$ , then the explanatory variable  $Y_{t+1}^e$  in equation (1) is simultaneously determined with the dependent variable  $Y_t$ , leading to biased regression estimates.*

Issue 2 indicates that causality in equation (1) is bidirectional, from  $Y_t$  to  $Y_{t+1}^e$  and vice versa. This necessitates finding an instrumental variable  $Z_t$  that influences  $Y_{t+1}^e$  independently of  $Y_t$ .

Formally, to address Issues 1 and 2, we introduce additional variation  $W_t$  to people's expectations in (2) as follows:

$$Y_{t+1}^e = b_0 + b_1 Y_t + W_t \tag{3}$$

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<sup>3</sup> $E[\cdot|\mathcal{F}_t]$  is the conditional expectation based on the information set  $\mathcal{F}_t$  at time  $t$ . This set includes knowledge that  $\{Y_t\}$  follows the process defined in equation (1).



where  $W_t$  is zero under FIRE. To solve Issue 1, we assume  $W_t \neq 0$  due to deviations from full information or rationality, and to solve Issue 2, we assume that some of the variation in  $W_t$  is observed and exogenous, allowing us to estimate a causal effect of  $Y_{t+1}^e$  on  $Y_t$ .

For illustrative purposes, we provide an example where  $W_t \neq 0$  due to noisy information about future shocks. In particular, consider agents receiving a signal about future shocks:

$$S_t = \epsilon_{t+1} + \eta_t \quad (4)$$

where  $\eta_t \stackrel{i.i.d}{\sim} N(0, \sigma_\eta^2)$  is independent of  $\epsilon_{t+1}$ . Agents are rational and solve a signal extraction problem, i.e.  $Y_{t+1}^e = E[Y_{t+1} | \mathcal{F}_t]$ , resulting in the following expression for  $W_t$  in (3):

$$W_t = E[\epsilon_{t+1} | \mathcal{F}_t] = \kappa(\epsilon_{t+1} + \eta_t), \quad \kappa = \frac{\sigma_\epsilon^2}{\sigma_\epsilon^2 + \sigma_\eta^2} \quad (5)$$

where  $\kappa$  is the Kalman gain. For example, signals about a potential future pandemic might initially lower output expectations. However, if the signal originates solely from noise  $\eta_t$ , actual output will only change if expectations have a real impact, specifically if  $\phi \neq 0$  in equation (1). Thus, the introduction of a signal regarding future shocks allows expectations to move independently of actual output at times, solving Issue 1 while adhering to the assumption of rational expectations. To address the simultaneity problem (Issue 2), one needs to observe some of the variation of the noise  $\eta_t$  in  $W_t$ , which can only move output  $Y_{t+1}$  if expectations matter. The next section introduces an instrumental variable approach to capture such variation and identify the effect of expectations.

## 3.2 Our Approach

To assess the impact of expectations  $Y_{t+1}^e$  on both current and future  $Y_{t+h}$  for  $h \geq 0$ , we use an instrumental variable  $Z_t$  that captures elements of expectations not influenced by  $Y_{t+h}$ , and estimate the following first-stage regression:

$$Y_{t+1}^e = \alpha_0 + \alpha_1 Z_t + \alpha_2 Y_{t-1} + \nu_t \quad (6)$$

The second stage regression for each impulse response function (IRF) horizon  $h$  is:

$$Y_{t+h} = \beta_0^{(h)} + \beta_1^{(h)} Y_{t+1}^e + \beta_2^{(h)} Y_{t-1} + \epsilon_{t+h}^{(h)} \quad h = 0, 1, 2, \dots \quad (7)$$

where the object of interest is the IRF:  $\{\beta_1^{(0)}, \beta_1^{(1)}, \dots, \beta_1^{(H)}\}$ .

The relevance and exclusion conditions for the instrumental variable are:

**A1** (Relevance Condition). *The instrument  $Z_t$  must be correlated with the explanatory variable  $Y_{t+1}^e$ , i.e.  $\text{Cov}(Z_t, Y_{t+1}^e) \neq 0$ .*

**A2** (Exclusion Restriction). *The instrument  $Z_t$  must not have any direct effect on the dependent variable  $Y_{t+h}$  apart from its effect through the explanatory variable  $Y_{t+1}^e$ , i.e.  $E[\epsilon_t^{(h)} | Z_t, Y_{t-1}] = E[\epsilon_t^{(h)} | Y_{t-1}]$  in (7).*

An ideal instrument fulfilling these conditions is an instrument whose change reflects exogenous expectation-shifting events that do not materialize, and any effects due to this change would originate from changes in expectations about the potential consequences of these events. In our case, when infectious diseases do not materialize, they have no direct real effects, yet the possibility of severe economic disruptions if they were to occur generates uncertainty and pessimism about future activity, making disease-related news a natural source of exogenous variation in output expectations. Therefore, an increase in such information ( $\uparrow Z_t$ ) might reduce expectations ( $\downarrow Y_{t+1}^e$ ), and if this information proves to be noise, then  $\epsilon_{t+1}$  remains unchanged, meaning any change in  $Y_t$  must be due to changes in expectations. For example, within the noisy information framework of (1), (3), (4), and (5), an ideal instrument  $Z_t$  would capture some of the fluctuations of the noise shock  $\eta_t$ , which can only affect the outcome  $Y_{t+h}$  through expectations  $Y_{t+1}^e$ .

Our interest extends beyond a single dependent variable to include multiple variables, such as instrumental and explanatory variables and other macro indicators. Consequently, instead of estimating the univariate first and second stages as in (6) and (7), we estimate a structural vector autoregression (SVAR) with an internal instrument (IV-SVAR), follow-

ing the methodology of [Romer and Romer \(2004\)](#), [Ramey \(2011\)](#), [Barakchian and Crowe \(2013\)](#), and [Jarociński and Karadi \(2020\)](#).

This IV-SVAR procedure consists of two steps. First, we estimate the impulse response functions (IRFs) to a shock to the instrument  $Z_t$  with a SVAR model of order  $p$ , represented as:

$$\mathbf{Y}_t = \gamma_0 + \Gamma_1 \mathbf{Y}_{t-1} + \cdots + \Gamma_p \mathbf{Y}_{t-p} + A\epsilon_t \quad (8)$$

Here,  $\mathbf{Y}_t = \begin{bmatrix} Z_t & Y_{t+1}^e & \mathbf{X}_t' \end{bmatrix}'$  is an  $n$ -dimensional vector of time series variables,  $\gamma_0$  is an  $n$ -dimensional vector of intercepts,  $\Gamma_l$  are  $n \times n$  matrices of autoregressive coefficients, and  $\epsilon_t$  are the structural VAR shocks, orthogonal to the lagged variables and to each other. The  $n \times n$  impact matrix  $A$  is derived via Cholesky decomposition of the residual covariance matrix, so the positioning of the instrument  $Z_t$  and the explanatory variable  $Y_{t+1}^e$  as first and second, before the other variables of interest  $\mathbf{X}_t$ , allows us to measure the effect of an exogenous shift in expectations  $Y_{t+1}^e$ , induced by the instrument.

The second step is to normalize the impulse of the first shock so that it decreases expectations  $Y_{t+1}^e$  by one. This normalization allows us to measure the causal effect of such an exogenous decrease in expectations on variables of interest. For an explanation of how this IV-SVAR model corresponds to the two-stage least squares method used in equations (6) and (7), refer to [Plagborg-Møller and Wolf \(2021\)](#).

## 4 Data and Estimation

The estimation of impulse response functions (IRFs) in response to an exogenous shift in expectations is conducted using the instrumental variable (IV) structural vector autoregression (SVAR) described in Section 3. The ordering of variables of the Cholesky-decomposed

VAR in (8) is as follows:

$$\mathbf{Y}_t = \begin{bmatrix} Z_t \\ Y_{t+1}^e \\ \mathbf{X}_t \end{bmatrix} = \begin{bmatrix} \text{Infectious Disease}_t \\ \text{Output Forecasts}_t \\ \text{Stock Prices}_t \\ \text{Interest Rate}_t \\ \text{Price Level}_t \\ \text{Output}_t \end{bmatrix} \quad (9)$$

This ordering uses the infectious disease index as an instrument for output forecasts, with the arrangement of the other variables not affecting the results.

The data used for the VAR model includes quarterly observations from the first quarter of 1985 until the fourth quarter of 2019 in the United States. This range is selected based on the availability of the infectious disease index and to avoid the influence of the COVID-19 pandemic, as it marks a distinct shift where the infectious disease index directly impacts the economy, not merely through expectations. The VAR is estimated with eight lags, equivalent to  $p = 8$  in (8), which is consistent with the typical approach of using two years of quarterly data to adequately reflect the dynamics in macroeconomic time series.

For the baseline estimation, we use the infectious disease index proposed by [Baker et al. \(2020\)](#) as the instrument  $Z_t$  to cause changes in economic agents' expectations. The original index is based on the major 11 U.S. newspapers on a daily basis, which we convert to a quarterly average to be consistent with the frequency of other variables used in the empirical model. The computation of the IDEMV tracker is done in two stages: First, they calculated the daily proportion of articles that contain a group of economic words, stock market terms, and market volatility phrases. The daily data was then multiplicatively rescaled to match the VIX value since 1985, yielding the equity market volatility (EMV) tracker. In the second stage, they chose EMV articles with at least one of the following terms: epidemic, pandemic, virus, flu, illness, Coronavirus, MERS, SARS, Ebola, H5N1 or H1N1. The IDEMV tracker is then constructed by multiplying the proportion of EMV

articles that contain one or more of these terms by the EMV tracker calculated in the first stage. This index captures information about potential pandemics, such as outbreak of infectious diseases that do not escalate into actual pandemics but may still be relevant for economics and financial markets.

Expectation variable is represented by output forecast, measured by 1-quarter ahead real GDP growth forecast, obtained from the Survey of Professional Forecasters organized by the Federal Reserve Bank of Philadelphia. Other data used in the estimation represent the U.S. business cycles, including the S&P 500 index, the three-month treasury bill rate, GDP deflator, and real GDP. All variables exhibit a unit root, as determined by the augmented [Dickey and Fuller \(1979\)](#) test, necessitating that they be differenced prior to estimation. The detailed description of the variables is outlined in [Appendix B](#).

## 5 Results

### 5.1 Baseline Results

[Figure 1](#) displays the impulse response functions (IRFs) to an exogenous shift in GDP forecast, which is caused by events related to infectious diseases. The size of the instrumental variable, the IDEMV, is normalized at a level that generates an initial decrease of GDP forecast by 1%.

An increase in perceived infectious-disease risk leads to a slightly-lagged deterioration in output expectations, reflecting slow information updating about whether such risks will evolve into a pandemic. Output forecasts reach a trough after roughly 2 quarters. Financial markets respond immediately, with stock prices fall on impact by about 5–6 percent, tightening financial conditions and depressing investment and consumption demand. Consistent with this contractionary demand channel, interest rates decline by up to 1 percentage point after 4–6 quarters, reflecting lower investment demand, increased precautionary saving, and endogenous policy accommodation. These forces translate into a gradual decline in real output, which reaches a peak contraction of about 0.49 percent, implying a pass-through of

roughly 50 percent from expectations to realized activity and highlighting that expectation shifts can have economically meaningful effects on aggregate fluctuations. As information about the disease scenario becomes more transparent and pandemic concerns fade, output expectations recover, contributing to a rebound and short-run overshooting in real output, while the price level remains persistently depressed, by close to 2 percent, due to sluggish price adjustment.

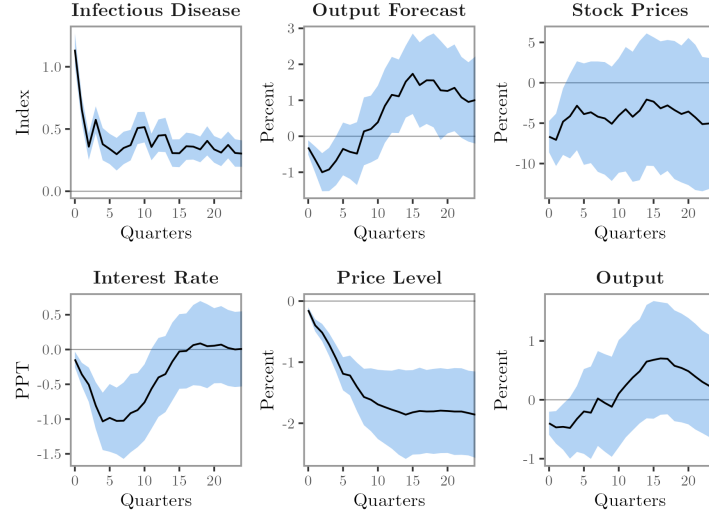


Figure 1: IRFs to an Exogenous Change in Expectations

NOTES – This figure shows the impulse response functions (IRFs) from an instrumental variable (IV) augmented structural vector autoregression (VAR) model with 8 lags, as outlined in equations (6), (7), (8), and (9). Solid lines represent the point estimates of the IRFs, illustrating the response of various macroeconomic indicators to an exogenous one-percent decline in output expectations over a 24-quarter horizon. The shaded areas indicate the 68% confidence intervals. The data used for the VAR estimation, shown in Figure B.1, have been differenced due to all six variables exhibiting a unit root, as identified by the augmented Dickey-Fuller test.

## 5.2 Robustness

To evaluate the robustness of the baseline results, Figures 2, 3, and 4 present IRFs from alternative model specifications. Figure 2 contrasts the benchmark instrumental variable, Baker, Bloom, Davis, and Kost’s (2019) IDEMV (grey shaded area), with an alternative model using Infectious Disease Frequency (IDF) as the instrument (dashed purple line). The IDF is constructed by converting the daily IDEMV index into a dummy variable (one if non-zero, zero otherwise) and averaging over each quarter, thereby removing IDEMV’s

dependency on the equity market.

Figure 3 displays the results where we incorporate various uncertainty indices as an additional control variable in the baseline VAR model: the CBOE Volatility Index (VIX), the Jurado, Ludvigson, and Ng’s (2015) macro uncertainty measure, and the EPU and EMV indices by Baker, Bloom and Davis (2016) and Baker et al. (2019).

In Figure 4, we compare the results from the baseline IV-SVAR to those from an instrumental-variable local projection (IV-LP) approach, as described by Jordà, Schularick and Taylor (2015), Leduc and Wilson (2017), Ramey and Zubairy (2018), and Jordà, Schularick and Taylor (2020).

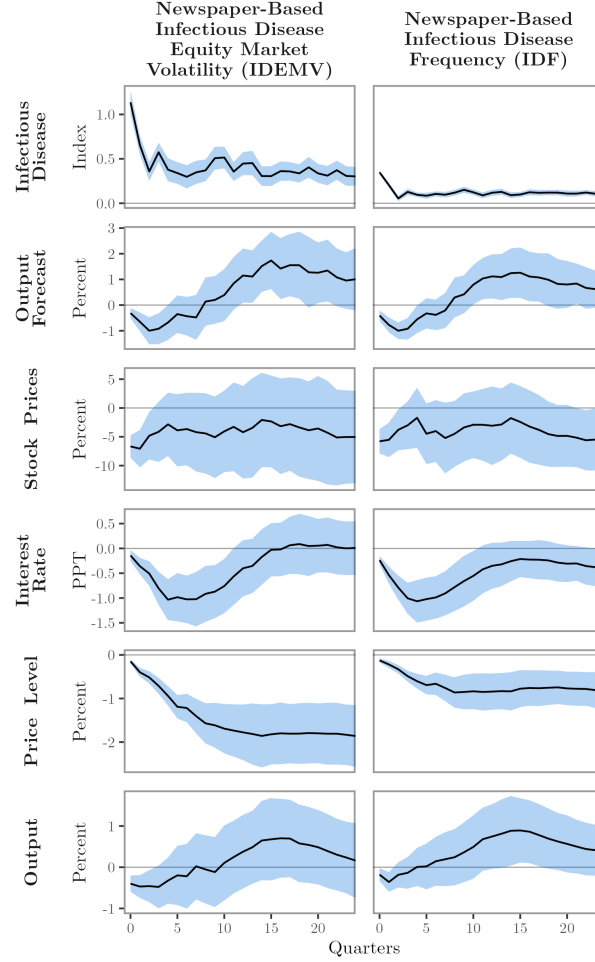


Figure 2: Robustness of IRFs to Instrumental Variable Selection

NOTES – The figure presents the impulse response functions (IRFs) for the benchmark IV-SVAR from Figure 1 alongside an alternative specification. The benchmark model, which uses Baker et al. (2020)’s IDEMV as the instrument (grey shaded area), is compared to a model using IDF as the instrument (dashed purple line). The IDF measures the frequency of days when infectious disease-related words such as “epidemic,” “pandemic,” “virus,” “flu,” “disease,” “coronavirus,” “mers,” “sars,” “ebola,” “H5N1,” and “H1N1” are mentioned in newspaper articles, and is constructed by converting the daily IDEMV index into a dummy variable, equalling one if non-zero and zero otherwise, and then averaging over each quarter. This approach removes IDEMV’s exposure to the equity market.



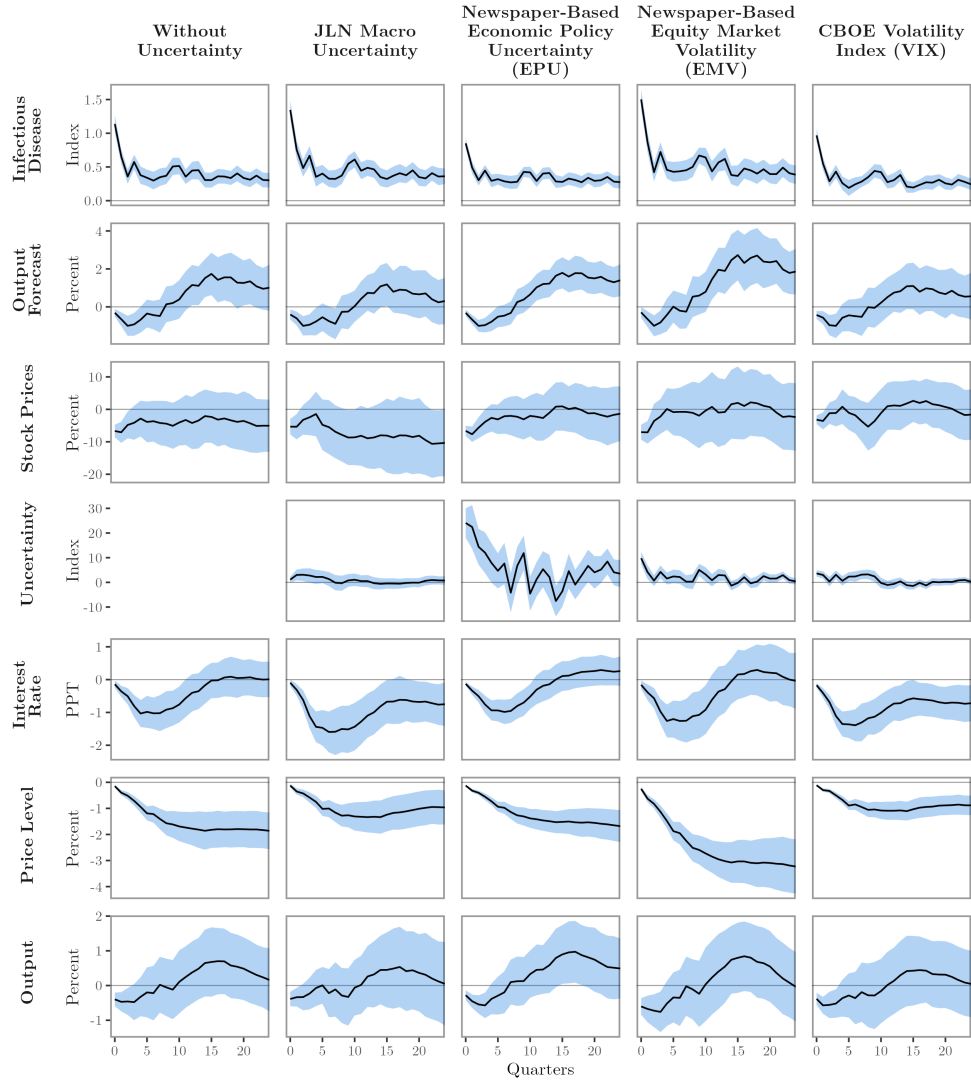


Figure 3: Robustness of IRFs to Uncertainty Proxies

NOTES – This figure presents a robustness check of impulse response functions (IRFs) using different proxies for uncertainty in the model: the CBOE Volatility Index (VIX), available under the “VIXCLS” ticker on the FRED database; Jurado, Ludvigson, and Ng’s (2015) measure of macro uncertainty, with data sourced from Sydney Ludvigson’s website at [www.sydneyludvigson.com](http://www.sydneyludvigson.com); and the EMV and EPU indices by Baker et al. (2019) and Baker, Bloom and Davis (2016) accessible at [www.policyuncertainty.com](http://www.policyuncertainty.com).

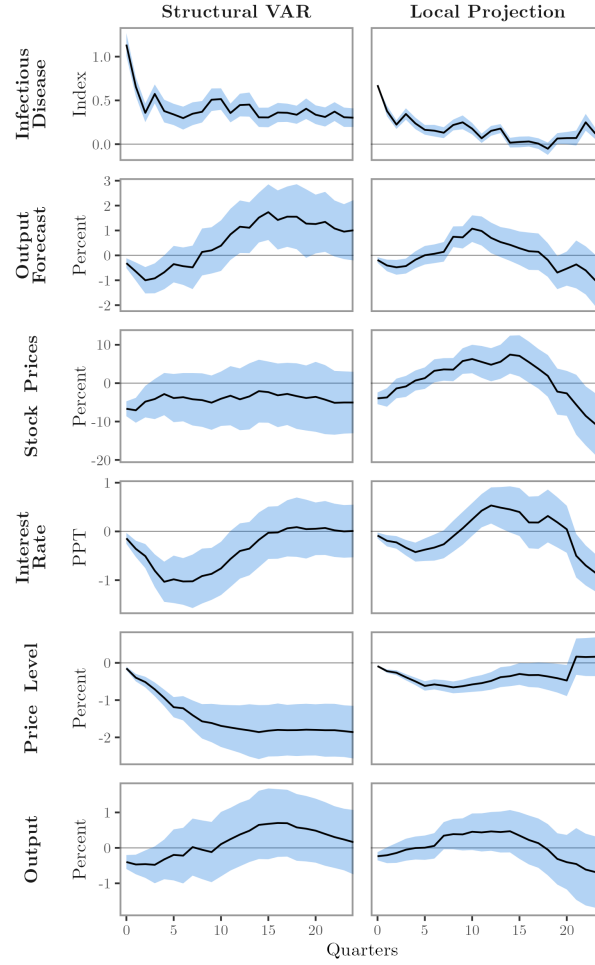


Figure 4: Robustness of IRFs to Model Selection

NOTES – This figure provides a robustness check for the impulse response functions (IRFs) by comparing the results from two different econometric approaches. The benchmark model (left panel) employs a structural vector autoregression (SVAR) with an internal instrument, where the IV is ordered first, as outlined in equations (6), (7), (8), and (9), and plotted in Figure 1. This approach follows the methodologies of [Ramey \(2011\)](#) and [Jarociński and Karadi \(2020\)](#). The alternative model (right panel) uses an instrumental-variable local projection (IV-LP) method, as described by [Jordà, Schularick and Taylor \(2015\)](#), [Leduc and Wilson \(2017\)](#), [Ramey and Zubairy \(2018\)](#), and [Jordà, Schularick and Taylor \(2020\)](#).

Across these diverse specifications, the IRFs demonstrate a remarkable alignment with the patterns of the benchmark model, suggesting that our estimates are robust to the choice of instrument, the inclusion of uncertainty measures, and modeling approach.

## 6 Conclusion

How people form expectations and how these expectations interact with the economy are key assumptions in business cycle models. In this paper, we empirically test the assumption that expectations matter using an instrumental variable framework, without imposing any predefined structure on how expectations are formed. An ideal instrument would shift expectations without directly impacting the economy, ensuring that any observed effects are solely caused by changes in expectations. We use news about infectious diseases that do not materialize into a pandemic as our instrument, providing a way to identify the impact of expectations on economic outcomes. Our findings show that the pass-through from real GDP expectations to actual real GDP is about 0.5, meaning that an exogenous decline in GDP forecasts by 1% can generate an actual decline in real GDP growth by 0.49%. Furthermore, we find these expectational effects to be transitory, and once expectations realign with reality, the expectation-induced movement is reversed.

Rather than proposing alternative behavioral assumptions, our approach complements the DSGE literature by focusing on empirical identification. While microfounded models typically assume that expectations influence economic outcomes, their effects are difficult to assess empirically using model-implied restrictions alone. Our contribution is to use an instrumental-variables strategy to directly estimate the causal impact of exogenous changes in expectations on macroeconomic outcomes, without attempting to validate or replace the underlying microfoundations.

For future research, exploring alternative instruments that shift expectations independently of current and future economic conditions will be essential. Examples of potential instruments that could be used to measure the effect of expectations using our approach include: (1) Information about revolutionary technologies that never come to fruition, prompting investment based on false expectations. (2) Political events that alter expectations about significant policy changes, which are later abandoned or not implemented. (3) Media-driven financial scares, such as sensationalized reporting on economic downturns

or recoveries not warranted by underlying economic fundamentals. (4) False alarms in international relations, such as unverified threats of conflict or peace agreements that shift expectations but do not result in any real action.

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## Appendix A Derivation of Equation (2) under FIRE

This appendix derives equation (2) under the assumption of full-information rational expectations (FIRE). We start from the data-generating process given in equation (1):

$$Y_t = \delta + \gamma Y_{t-1} + \phi Y_{t+1}^e + \varepsilon_t, \quad (\text{A.1})$$

where  $\varepsilon_t \stackrel{i.i.d.}{\sim} N(0, \sigma_\varepsilon^2)$  represents an independent shock. Under FIRE, agents form expectations according to

$$Y_{t+1}^e = \mathbb{E}_t[Y_{t+1}], \quad (\text{A.2})$$

and, by the full-information assumption,  $\mathbb{E}_t[\varepsilon_{t+1}] = 0$ .

Shifting equation (1) forward by one period yields

$$Y_{t+1} = \delta + \gamma Y_t + \phi Y_{t+2}^e + \varepsilon_{t+1}. \quad (\text{A.3})$$

Taking expectations conditional on the information set at time  $t$  gives

$$\mathbb{E}_t[Y_{t+1}] = \delta + \gamma Y_t + \phi \mathbb{E}_t[Y_{t+2}^e]. \quad (\text{A.4})$$

Under rational expectations,  $\mathbb{E}_t[Y_{t+2}^e] = \mathbb{E}_t[Y_{t+2}]$ , so expected output satisfies

$$Y_{t+1}^e = \delta + \gamma Y_t + \phi \mathbb{E}_t[Y_{t+2}]. \quad (\text{A.5})$$

To solve for expectations, conjecture a linear forecasting rule of the form

$$Y_{t+1}^e = b_0 + b_1 Y_t. \quad (\text{A.6})$$

Using this conjecture, expected output two periods ahead is given by

$$\mathbb{E}_t[Y_{t+2}] = b_0 + b_1 \mathbb{E}_t[Y_{t+1}] = b_0 + b_1(b_0 + b_1 Y_t). \quad (\text{A.7})$$



Substituting this expression into equation (A.5) yields

$$b_0 + b_1 Y_t = \delta + \gamma Y_t + \phi [b_0 + b_1(b_0 + b_1 Y_t)]. \quad (\text{A.8})$$

Matching coefficients on the constant and  $Y_t$  terms implies

$$b_0 = \delta + \phi b_0(1 + b_1), \quad (\text{A.9})$$

$$b_1 = \gamma + \phi b_1^2. \quad (\text{A.10})$$

These conditions characterize the fixed point of the rational expectations forecasting rule. Thus, under full-information rational expectations, expected output is a linear function of current output,

$$Y_{t+1}^e = b_0 + b_1 Y_t, \quad (\text{A.11})$$

which corresponds to equation (2) in the main text.

## Appendix B Data appendix

The variables in the baseline IV-SVAR model, shown in Figure B.1, are defined as follows:

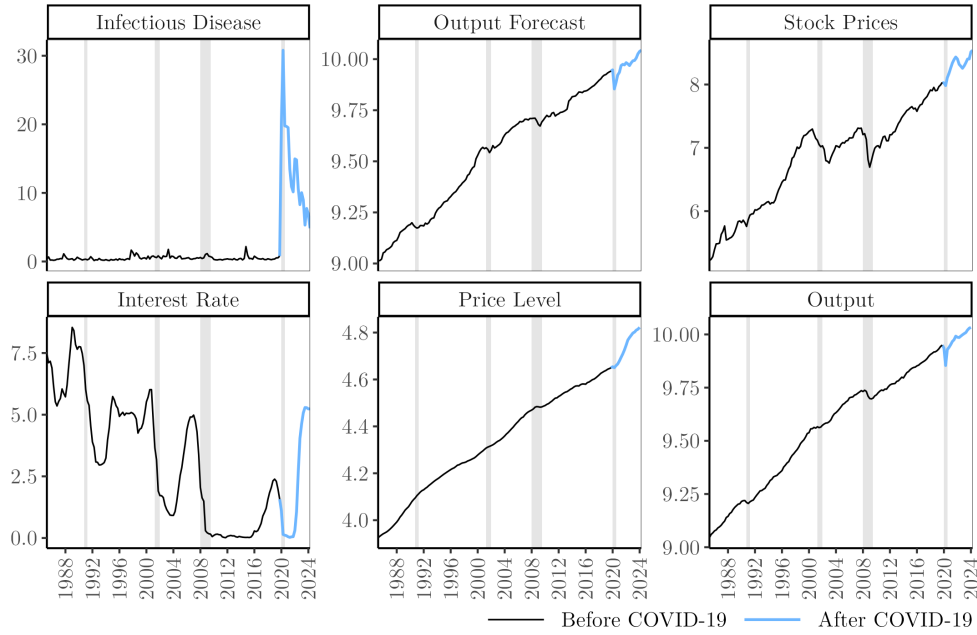


Figure B.1: Data

NOTES – This figure displays the economic indicators used in the VAR estimation as outlined in equations (8) and (9). It includes the quarterly average of the daily newspaper-based infectious disease index developed by Baker et al. (2019); the natural logarithm of the median one-quarter ahead real GDP forecast from the Philadelphia Fed’s Survey of Professional Forecasters; the natural logarithm of the S&P 500 index from Yahoo Finance; the three-month Treasury Bill rate from the Federal Reserve Board; the natural logarithm of the implicit GDP price deflator reported by the Bureau of Economic Analysis (BEA); and the natural logarithm of real GDP, also reported by the BEA. The gray shades represent recession periods as identified by the National Bureau of Economic Research (NBER).

1. *Infectious Disease* is the quarterly average of the daily newspaper-based infectious disease index developed by Baker et al. (2019), accessible at [www.policyuncertainty.com](http://www.policyuncertainty.com). For additional details on the methodology used to create newspaper-based indices, refer to Baker, Bloom and Davis (2016); Baker et al. (2019); Baker et al. (2020).
2. *Output Forecast* represents the natural logarithm of the median forecast for real GDP, sourced from the Philadelphia Fed’s Survey of Professional Forecasters (SPF).

3. *Stock Prices* refers to the natural logarithm of the S&P 500 index, sourced from Yahoo Finance under the ticker symbol “^GSPC”.
4. *Interest Rate* denotes the three-month Treasury Bill rate sourced from the secondary market on a discount basis, provided by the Board of Governors of the Federal Reserve System and available in the Federal Reserve Economic Data (FRED) database under the identifier “TB3MS”.
5. *Price Level* is the natural logarithm of the implicit GDP price deflator, reported by the Bureau of Economic Analysis, available on FRED under the “GDPDEF” symbol.
6. *Output* represents the natural logarithm of real gross domestic product (GDP), reported by the Bureau of Economic Analysis and accessible on FRED under the “GDPC1” symbol.